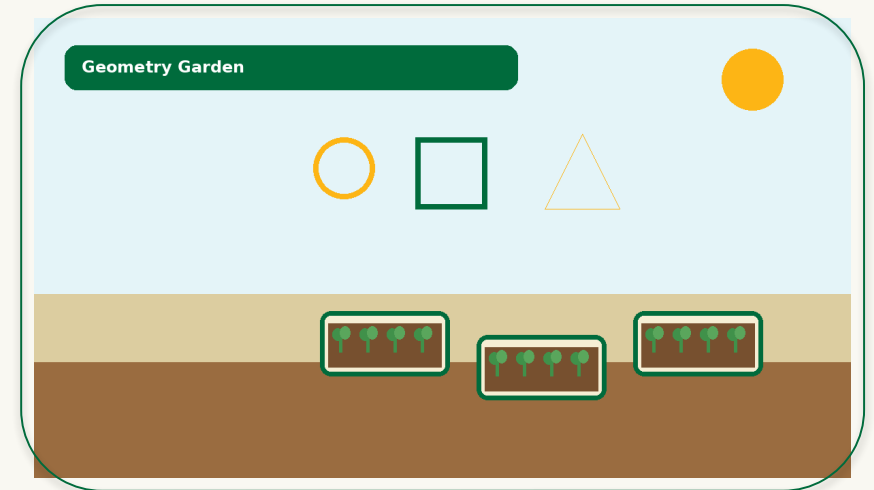


Geometry Garden

Applied Mathematics Through Garden Design

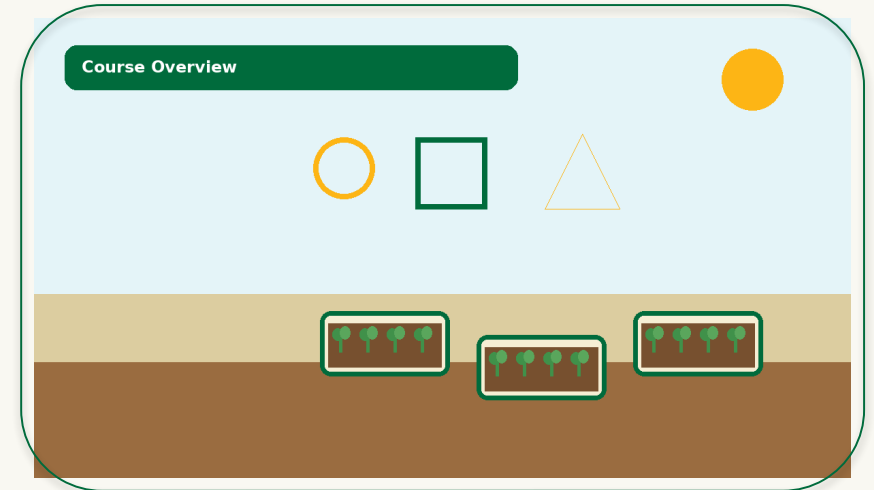
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Course emphasis: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden system and its educational purpose; teach the ten technical topics at Primary 4-6 level; connect mathematics/science/agriculture to practical work; supervise measurements and garden tasks safely; maintain usable records; guide pupils from observation toward evidence-based conclusions.
- Core materials: teacher guide, crop plan, child-safe tools, watering equipment, labels, clipboards, rulers/tapes, scales or thermometers where relevant, observation/data sheets, clean containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson/measurement or investigation plan.



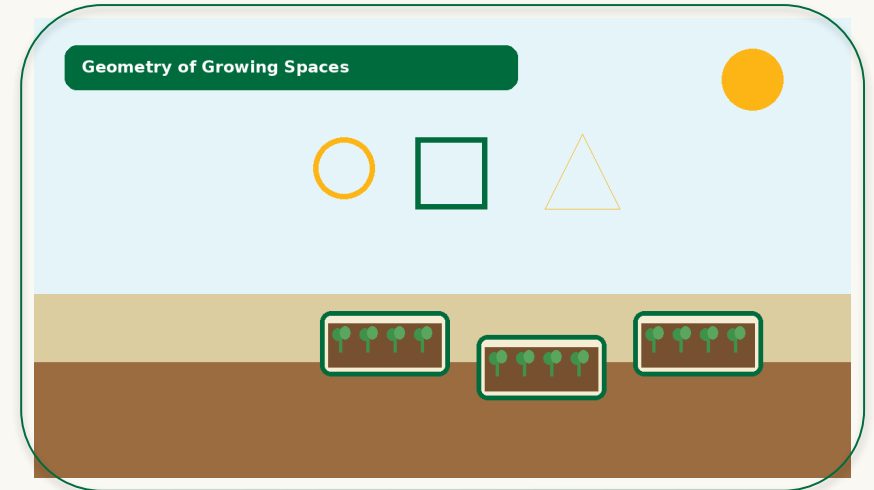
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

Geometry of Growing Spaces

Teacher knowledge and practical application

- Use real beds, paths and containers to identify two-dimensional shapes and spatial arrangements.
- Connect mathematical vocabulary to physical growing spaces.
- Distinguish the outline of a bed from the three-dimensional structure around it.
- Ask pupils to describe properties rather than only name shapes.
- Use garden maps to move between real space and drawn representation.



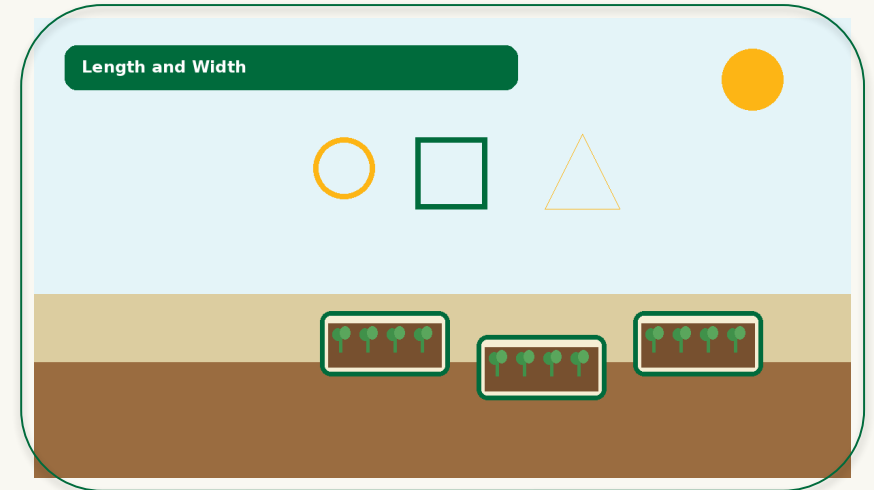
TEACHER PRACTICE

- Prepare a geometry walk using four garden features.

Length and Width

Teacher knowledge and practical application

- Define length and width consistently for the feature being measured.
- Demonstrate correct placement of a tape or ruler from zero point to endpoint.
- Estimate before measuring, then compare estimate with result.
- Record both value and unit.
- Repeat measurements when readings disagree.



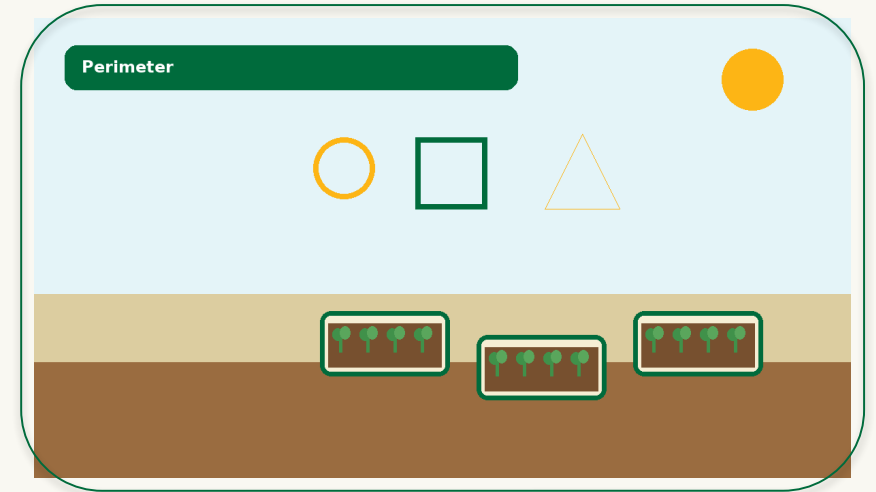
TEACHER PRACTICE

- Prepare a paired length-and-width measurement activity.

Perimeter

Teacher knowledge and practical application

- Introduce perimeter as distance around a boundary.
- Measure sides of rectangular or other manageable beds and add side lengths.
- Use a walking boundary to reinforce the concept before calculation.
- Keep measurement paths safe and outside planted areas.
- Discuss why perimeter matters for edging, fencing or border materials.



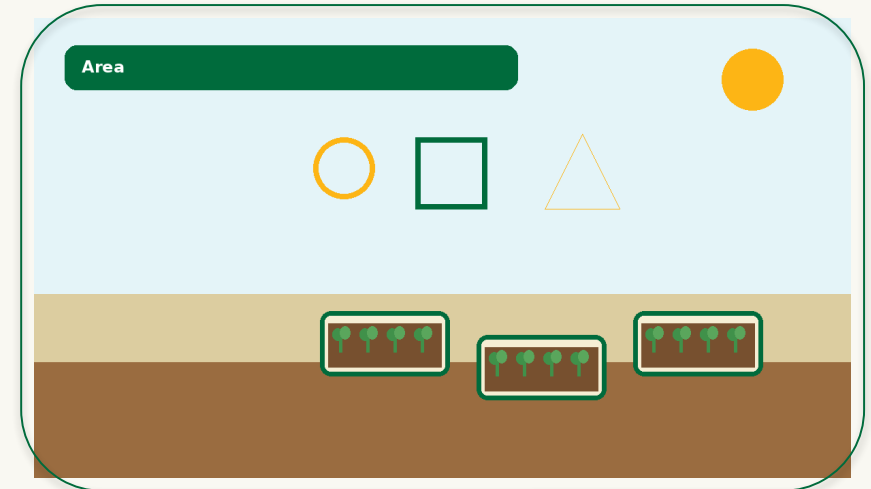
TEACHER PRACTICE

- Calculate the perimeter of one real garden bed.

Area

Teacher knowledge and practical application

- Introduce area as the amount of surface inside a boundary.
- Use square units and grid representations before formulas where appropriate.
- Calculate rectangular bed area using length x width after accurate measurement.
- For irregular spaces, use simple grid estimation rather than false precision.
- Connect area to planning planting space and material quantities.



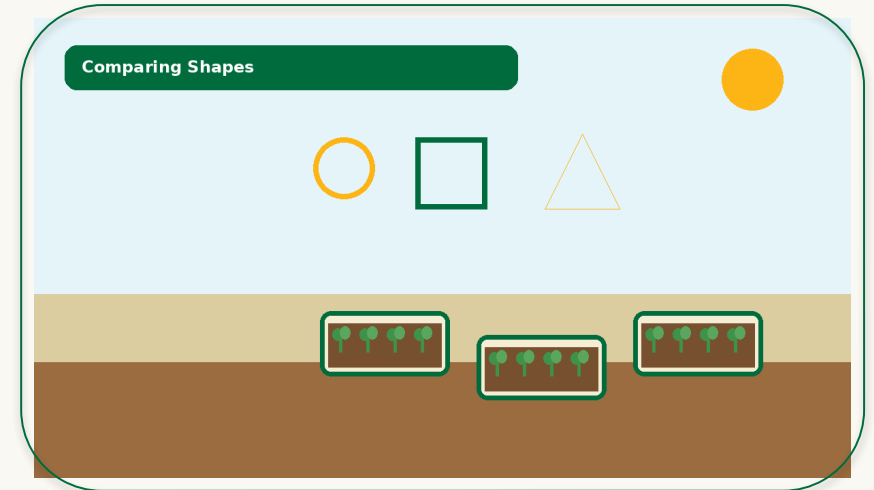
TEACHER PRACTICE

- Measure and calculate the area of one rectangular bed.

Comparing Shapes

Teacher knowledge and practical application

- Compare shapes by sides, angles/corners, perimeter and area without mixing attributes.
- Show that shapes with the same area can have different perimeters.
- Use physical garden examples and scaled drawings.
- Ask pupils to state the basis of each comparison.
- Avoid assuming visually larger means mathematically larger without measurement.



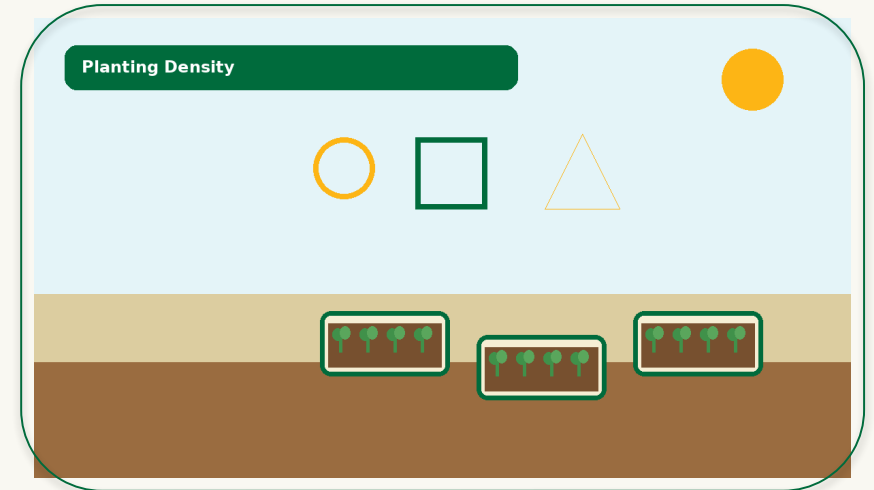
TEACHER PRACTICE

- Prepare a two-bed comparison using at least two attributes.

Planting Density

Teacher knowledge and practical application

- Connect area to number of plants placed within a bed.
- Calculate simple plants-per-area examples using actual counts.
- Explain that recommended spacing limits how many plants should occupy the area.
- Compare equal-area beds with different planting arrangements.
- Keep agricultural requirements primary when designing mathematical examples.



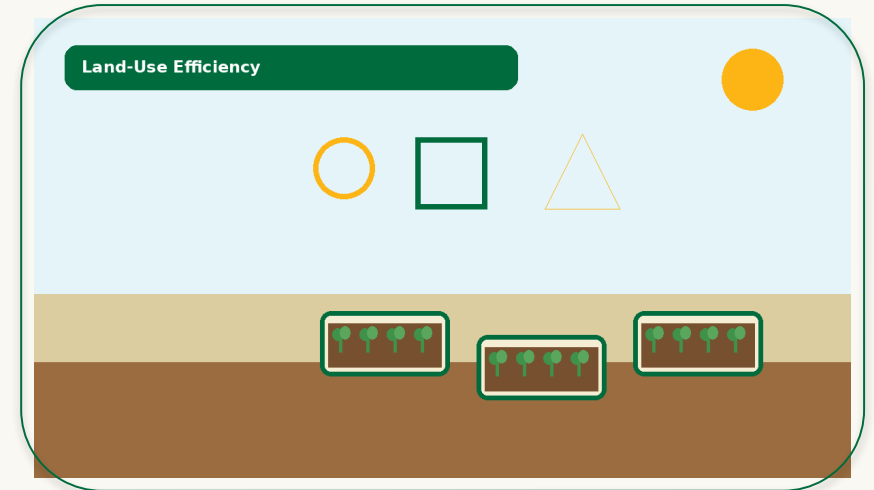
TEACHER PRACTICE

- Calculate density for one planted bed.

Land-Use Efficiency

Teacher knowledge and practical application

- Introduce the idea as how effectively available garden space is organized for the stated purpose.
- Compare productive bed area with paths and non-growing areas using simple measurements.
- Explain that paths are necessary for access, so maximum planted area is not automatically best.
- Use percentages only where appropriate to the grade level.
- Discuss trade-offs between access and growing area.



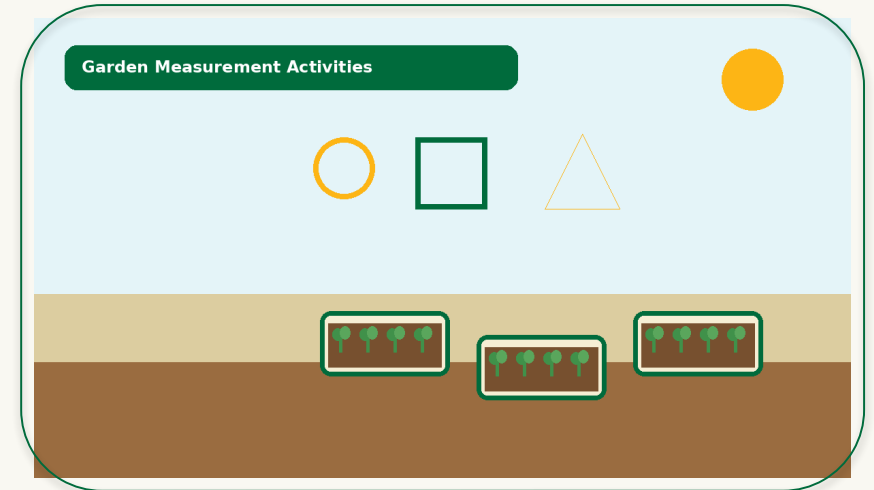
TEACHER PRACTICE

- Create a simple bed-versus-path area comparison.

Garden Measurement Activities

Teacher knowledge and practical application

- Use measurement stations for bed dimensions, path width, container height and spacing.
- Assign roles: measurer, checker and recorder.
- Require units on every recorded value.
- Repeat selected measurements for quality control.
- Keep pupils away from unstable structures and active equipment.



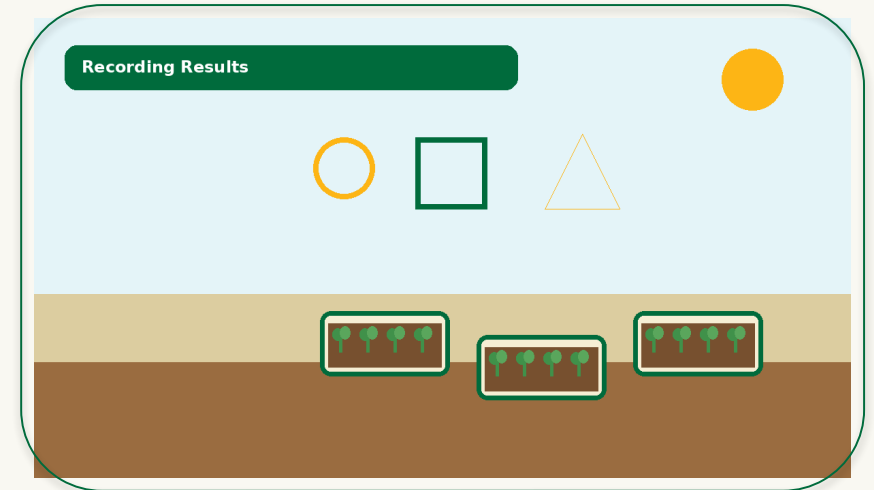
TEACHER PRACTICE

- Plan four measurement stations for a 40-minute lesson.

Recording Results

Teacher knowledge and practical application

- Use tables with feature, measurement, unit and date.
- Show how neat records reduce calculation errors.
- Check calculations against the original measurements.
- Use simple diagrams with dimensions.
- Separate measured values from calculated values.



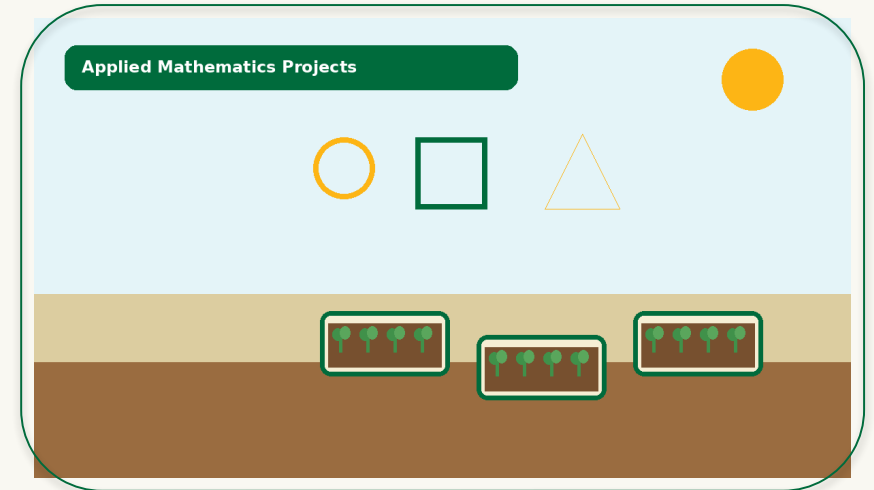
TEACHER PRACTICE

- Prepare a garden measurement table and sketch.

Applied Mathematics Projects

Teacher knowledge and practical application

- Design small projects such as planning a new bed, estimating edging length or comparing usable area.
- Require a question, measurements, calculation and short explanation.
- Use realistic garden constraints such as paths and crop spacing.
- Assess mathematical reasoning as well as numerical answers.
- Encourage pupils to verify results in the physical garden.



TEACHER PRACTICE

- Completion task: design a garden-mathematics mini-project.